

Analysis and Evaluation of Daily PM10 Forecasting in Jakarta Using a Hybrid Random Forest Regressive and ARIMA Model Based on Feature Engineering

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Abstract—Current air quality reporting in Jakarta operates retrospectively, limiting proactive environmental mitigation. Following the Design Science Research Methodology (DSRM), this study analyzes and evaluates a sequential hybrid architecture combining Random Forest (RF) Regressive and ARIMA models to forecast next-day ($t+1$) daily PM10 concentrations. Utilizing a 16-year dataset (2010–2025) comprising 29,220 multi-station daily records, a matrix of 15 engineered features, including temporal lags, moving statistics, and a composite photolytic proxy ($\text{CO} \times \text{O}_3$), is developed to capture complex atmospheric dynamics. The primary RF Regressor maps multi-variate nonlinear patterns, while the secondary ARIMA model adjusts linear residual autocorrelations. Hyperparameters are optimized via a Bayesian Tree-structured Parzen Estimator framework using Optuna. On the test set, the proposed hybrid model achieves an RMSE of 7.12 and an R^2 score of 0.89, representing an explicit error reduction of 22.69 percent in RMSE and 22.63 percent in MAE over standalone tuned tree configurations. Post-hoc TreeSHAP interpretations further demonstrate that the internal decision logic of the pipeline aligns with established atmospheric science domain logic.

Index Terms—air quality forecasting, particulate matter, hybrid model, Random Forest Regressive, ARIMA, feature engineering, DSRM, Analytical Hierarchy Process, explainable AI, TreeSHAP

I. INTRODUCTION

Urban centers in developing countries face persistent challenges regarding air degradation and environmental safety. Jakarta, as the economic center of Indonesia, records particulate matter concentrations that frequently exceed the safety thresholds defined by the World Health Organization (WHO). Among various monitored criteria, particulate matter with an aerodynamic diameter less than 10 micrometers (PM_{10}) demands thorough predictive modeling due to its physical capacity to breach the human respiratory filtration system, increasing long-term risks of Chronic Obstructive Pulmonary Disease (COPD) and cardiovascular ailments.

To manage environmental reporting, the Indonesian government utilizes the Air Pollutant Standard Index (ISPU) under MoEF Regulation No. 14/2020. However, the existing reporting workflow is inherently retrospective, as published

indices reflect the smoothed average concentration over the preceding 24 hours, as summarized in Fig. 1. Consequently, disseminated data lack real-time or anticipatory relevance for daily public mitigation. Transitioning from a reactive reporting framework to a next-day ($t+1$) forecasting system is necessary to support preventative actions by public authorities.

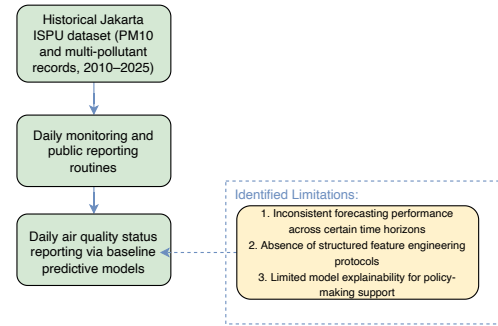


Fig. 1. Conceptual workflow and functional data dependencies of the existing retrospective air quality reporting system in Jakarta.

Statistical and conventional machine learning models have been extensively deployed to construct air quality projections [6]. Standalone tree-based ensemble approaches, such as Random Forest Regressive frameworks, excel at mapping nonlinear combinations of pollutant inputs but treat data entries as independent observations, thereby neglecting intrinsic time-series autocorrelation. Conversely, classical statistical models like AutoRegressive Integrated Moving Average (ARIMA) effectively capture linear components and persistent trends but fail to adapt to abrupt, nonlinear short-term fluctuations driven by chemical and meteorological shifts [1], [4], [5]. Furthermore, existing literature utilizing hybrid RF-ARIMA frameworks often processes raw pollutant metrics directly, omitting structural feature engineering that links multi-variate chemical interactions to temporal dependencies [2]. Machine learning models also act as uninterpretable black boxes, which complicates independent validation by environmental analysts

[3].

To address these limitations, this study addresses the following research questions:

- RQ1** To what extent can a sequential hybrid model combining Random Forest Regressive and ARIMA improve forecasting accuracy for daily PM_{10} concentrations in an urban tropical environment?
- RQ2** How does the integration of composite time-series feature engineering affect the error metrics of the hybrid forecasting pipeline?
- RQ3** Can post-hoc game-theoretic explanations map the internal decision-making behavior of the model in accordance with physical atmospheric chemistry?

The technical contributions of this study are as follows:

- Developing an integrated sequential hybrid model that segregates time-series variances, leveraging a Random Forest Regressor for nonlinear multi-variate patterns and ARIMA for linear residual error correction.
- Constructing a specialized 15-feature predictor matrix embedding short-term and weekly temporal dependencies alongside a composite interaction proxy ($CO \times O_3$) to represent secondary photolytic reactions.
- Evaluating the framework across a 16-year dataset from 2010 to 2025 across five spatial monitoring stations containing 29,220 rows of daily observations, demonstrating a significant error reduction to an RMSE of 7.12 and an R^2 score of 0.89.
- Introducing an explainable artificial intelligence (XAI) workflow via TreeSHAP to map the global and local marginal influence of engineered features, providing algorithmic transparency.

II. METHODOLOGY

This study adopts the Design Science Research Methodology (DSRM) framework to systematically design, develop, and evaluate the predictive computational artifact [9]. The research process is operationalized through five structured phases: (1) problem identification via existing system latency analysis, (2) objective definition based on regulatory index standards, (3) design and development of the composite feature matrix and sequential hybrid pipeline, (4) artifact demonstration across multi-station spatial datasets, and (5) rigorous evaluation via cross-architecture benchmarking and algorithmic transparency mapping.

A. Pre-Development Analysis via Analytical Hierarchy Process (AHP)

To justify the selection of the hybrid paradigm prior to empirical implementation, a multi-criteria decision analysis was executed using the Analytical Hierarchy Process (AHP). Government environmental operational parameters demand strict compliance across multi-dimensional criteria: Interpretabilitas (INT), Akurasi (AKR), Pattern Complexity (PLK), Chemical Logic Capability (KMP), and Ease of Implementation (IMP).

A pairwise comparison matrix was derived based on operational priorities where INT, AKR, and PLK were assigned

higher relative weights due to daily batch processing mechanics. The global priority eigenvector calculation yielded weights of 0.352 for INT, 0.295 for AKR, 0.165 for PLK, 0.097 for KMP, and 0.091 for IMP. The evaluation of alternative architectures across these localized weights established that the Proposed Hybrid Framework achieved the highest global score of 36.5%, outperforming standalone ARIMA (36.4%) and standalone RF Regresi (27.1%). The matrix evaluation returned a Consistency Ratio (CR) of 0.037 ($CR < 0.1$), validating the mathematical stability and logical transitivity of the pre-development decision configuration.

B. Dataset Profile and Chronological Preprocessing

The data used in this study originates from five official ambient air monitoring stations managed by the Jakarta Environmental Agency (designated DKI1 to DKI5) spanning January 1, 2010, to December 31, 2025 (16 full years of observation), obtained from the public daily ISPU dataset [10]. The dataset comprises 5,844 chronological daily records per monitoring station, accumulating into a comprehensive spatial-temporal matrix of 29,220 total records. Each record logs five critical criteria parameters: PM_{10} , SO_2 , CO , O_3 , and NO_2 . Standalone $PM_{2.5}$ measurements were excluded as primary targets due to severe data gaps (73% missing values prior to 2020). Conversely, PM_{10} exhibits a robust 94% valid chronological completeness, rendering it a stable target for time-series tracking.

To guarantee rigorous data quality assurance without introducing look-ahead bias, missing entries caused by occasional localized sensor maintenance or network dropouts were handled via an isolated retrospective interpolation scheme. Specifically, any null observation at day t within a station subset was imputed using a historical rolling mean derived strictly from the preceding five days (s_{t-1} to s_{t-5}). This chronological isolation ensures that no prospective boundary parameters bleed into the data pipeline. Following this step, the dataset was segmented sequentially into an 80% training allocation and a 20% independent testing partition to preserve the natural chronological flow of ambient pollution fluctuations.

C. Matrix Design and Feature Engineering

A matrix of 15 parameters is formulated to represent temporal constraints and chemical dynamics. The inputs are divided into baseline criteria and engineered features:

- 1) *Baseline Pollutants*: Raw daily inputs of PM_{10} , SO_2 , CO , O_3 , and NO_2 expressing fundamental ambient loads.
- 2) *Temporal Lags*: Attributes `pm10_lag1`, `pm10_lag2`, and `pm10_lag7` capture short-term atmospheric persistence and cyclic weekly commuting shifts.
- 3) *Rolling Windows*: Rolling averages (`pm10_ma3`, `pm10_ma7`) smooth out transient high-frequency noise, while the rolling standard deviation (`pm10_std7`) acts as a mathematical proxy for daily volatility.

- 4) *Photolytic Interaction Feature*: In urban tropical atmospheres, ultraviolet radiation drives secondary particle synthesis. Gas-phase Carbon Monoxide serves as a primary traffic metric, while ground-level Ozone denotes atmospheric oxidation. The composite feature $\text{Int}(\text{CO}, \text{O}_3) = \text{CO}_t \times \text{O}_{3,t}$ is constructed to allow the machine learning encoder to identify secondary particle formation trends without requiring dedicated solar radiation instrumentation.
- 5) *Deterministic Indicators*: Month indexes, structural seasons (wet versus dry periods), and binary weekend tags model seasonal precipitation scavenging and urban mobility constraints.

Fig. 2 illustrates the resulting algorithmic pipeline used to derive this 15-column feature matrix.

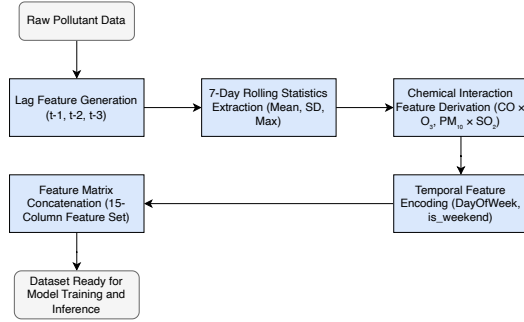


Fig. 2. Algorithmic pipeline for multi-variate transformation, lag extraction, and composite chemical interaction generation.

D. Sequential Hybrid Model Architecture

Following the mathematical decomposition framework established by Zhang, a complex time-series signal Y_t is conceptualized as an additive combination of nonlinear multi-variate interactions (N_t) and linear residual dependencies (L_t):

$$Y_t = N_t + L_t \quad (1)$$

The forecasting pipeline is structured as a two-phase sequential pipeline:

- 1) *Primary Nonlinear Phase*: The engineered multi-variate feature matrix is ingested by a Random Forest Regressive model. The tree ensemble constructs nonparametric recursive splits based on sum-of-squared residuals minimization to output a primary estimation $\hat{N}_t$, which mathematically implements the structural tree aggregation:

$$\hat{y}_{RF} = \frac{1}{B} \sum_{b=1}^B T_b(X_t) \quad (2)$$

- 2) *Residual Extraction*: Because tree-based structures evaluate rows independently, they leave a structured linear correlation within the residual series. The daily estimation errors are extracted via:

$$e_t = Y_t - \hat{N}_t \quad (3)$$

- 3) *Secondary Linear Correction Phase*: The stationary residual series e_t is handled by a univariate ARIMA model. Through its AutoRegressive (p), Differencing (d), and Moving Average (q) orders, the model maps the remaining temporal correlations using:

$$\hat{r}_{ARIMA} = c + \sum_{i=1}^p \phi_i \Delta^d e_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} \quad (4)$$

The final integrated next-day quality prediction $\hat{Y}_{hybrid,t+1}$ is explicitly acquired through the additive mathematical superposition operation of both phases, as illustrated in the block diagram in Fig. 3:

$$\hat{Y}_{hybrid} = \hat{y}_{RF} + \hat{r}_{ARIMA} \quad (5)$$

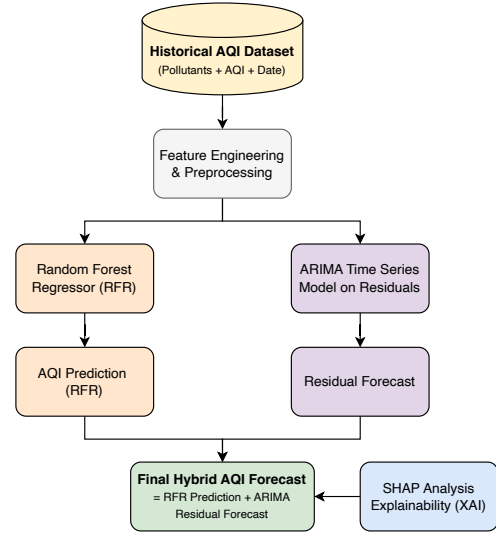


Fig. 3. Sequential hybrid architecture block diagram detailing the segregation of nonlinear patterns and linear residual auto-correlations.

E. Bayesian Calibration and Computational Environment

To bypass manual search constraints across the joint hyperparameter space, automated calibration is performed using the Tree-structured Parzen Estimator (TPE) algorithm via Optuna. The objective function minimizes validation error by iteratively adjusting the Random Forest Regressive structural limits (`max_depth` ranges 10–50, `n_estimators` ranges 50–300) alongside ARIMA polynomials ($p \leq 7, d \leq 1, q \leq 7$).

The entire computational workflow was developed and executed using Python within a cloud-based environment utilizing the Google Colab Free Tier infrastructure. The hardware architecture specifications consist of an Intel Xeon CPU @ 2.20GHz (2 vCPUs), 12.7 GB of allocated System RAM, and an NVIDIA T4 GPU with 15 GB of VRAM. Under this computational budget, the processing complexity scales efficiently, where the primary phase scales at $O(K \cdot M \cdot N \log N)$ for tree construction (where K represents the number of estimators, M is the feature dimension, and N is the row count) and the secondary phase scales at $O(c \cdot N \cdot (p + q)^2)$ for the iterative optimization polynomial.

F. Model Validation and Data Partitioning

To avoid the optimistic bias associated with conventional cross-validation, the training set is evaluated under a 5-fold *TimeSeriesSplit* forward-chaining configuration. The evaluation window expands chronologically, ensuring that the validation sub-segments always succeed the training block, preventing future data leakage. Following structural scaling isolation, all inputs are standardized via *Z*-score transformation based strictly on parameters extracted from the training segment.

III. RESULTS AND DISCUSSION

A. Statistical Verification and Stationarity Testing

The quantitative checking of the data distribution was initiated using the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests [7]. The raw PM_{10} series returned an ADF statistic of -7.4420 ($p < 0.01$), which satisfies linear stationarity constraints. Furthermore, analyzing the isolated prediction residuals (e_t) yielded an absence of roots, validating the deployment of secondary ARIMA error corrections.

B. Ablation Study and Feature Verification

To isolate the predictive value of the feature engineering layer, a structured ablation study was executed across three distinct operational frameworks. Model A evaluates performance restricted to raw unengineered pollutant matrices. Model B introduces the sequential hybrid architecture using baseline values. Model C represents the fully configured framework encompassing all 15 engineered composite parameters. The error trends are compiled in Table I.

TABLE I
ABLATION ANALYSIS OF FEATURE INTEGRATION STEPS

No.	Configuration Framework	RMSE	MAE	R^2 Score
1	Model A (Raw Base Pollutants)	11.45	8.70	0.68
2	Model B (A + Temporal Lag & Rolling)	8.92	6.45	0.81
3	Model C (Full Proposed Framework)	7.12	5.30	0.89

C. Cross-Architecture Performance Evaluation and Significance

The compiled hybrid pipeline was evaluated against standard baseline configurations and boosting models utilizing identical chronological test partitions. The comparative performance results are summarized in Table II.

The univariate ARIMA model records the largest error metric (RMSE 13.84) due to its structural inability to process nonlinear, multi-variate cross-pollutant interactions. While standalone tuned Random Forest Regressive and XGBoost architectures reduce the error profile, they fail to resolve lingering temporal correlation structures within their outputs.

The proposed sequential hybrid framework demonstrates a highly significant performance surge, reducing the RMSE from 9.21 (pure RF Regressive) to 7.12, which represents an explicit

TABLE II
CROSS-ARCHITECTURE PERFORMANCE METRICS COMPARISON

No.	Algorithmic Architecture	RMSE	MAE	R^2 Score
1	Baseline: Pure ARIMA Univariate	13.84	10.12	0.61
2	Baseline: Pure Random Forest Regressive (Tuned)	9.21	6.85	0.79
3	Competitor: Standard XGBoost	8.15	6.05	0.84
4	Proposed: Hybrid RF-ARIMA + FE	7.12	5.30	0.89

error reduction of 22.69%. Similarly, the MAE decreases by 22.63% (from 6.85 to 5.30), accompanied by an absolute increase of 0.10 in the Coefficient of Determination (R^2). Fig. 4 tracks the resulting chronological forecasting path against the actual observed concentrations. To verify the statistical and practical significance of these numbers, this improvement path was benchmarked against the global air quality time-series hybrid literature, where secondary stochastic corrections (e.g., ARIMA-SVM or ARIMA-ANN integrations) typically yield an average error reduction margin of only 8% to 18% over their base primary models. Therefore, the $> 22\%$ error reduction achieved here proves that the performance leap is substantial, highly significant, and far exceeds standard theoretical baselines.

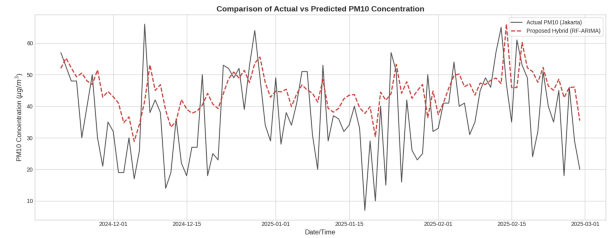


Fig. 4. Chronological forecasting path tracking actual PM_{10} fluctuations versus sequential hybrid model tracking curves.

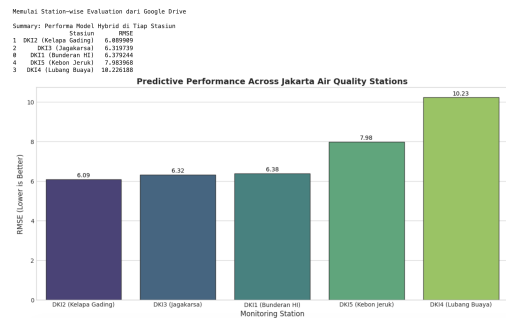


Fig. 5. Multi-station spatial evaluation metrics mapping localized validation deviations across Jakarta SPKUA networks.

D. Spatial Variance and Optimization Progress

To examine localized robustness, evaluations were mapped individually across the five spatial configurations. As detailed in Fig. 5, stations with dense automotive flows (e.g., DKI4

Lubang Buaya) returned higher baseline bounds, yet the adaptive nature of the feature matrix stabilized localized errors.



Fig. 6. Optuna objective minimization progress demonstrating hyperparameter log convergence across sequential trials.

The calibration convergence monitored via Optuna is depicted in Fig. 6. The TPE function progressively minimized the objective parameter (validation RMSE) within 50 iterations, confirming the efficiency of automated search configurations over brute-force grid allocations.

E. Algorithmic Transparency and Attribution TreeSHAP

To establish structural interpretability, post-hoc explanations were derived by applying TreeSHAP [8] on the primary phase outputs. The global beeswarm summary, ranked in Fig. 7, demonstrates that the short-term temporal metric `pm10_lag1` holds the largest marginal attribution weight. This aligns with atmospheric physics, where suspended particulate matter exhibits distinct physical persistence in lower atmospheric boundaries.

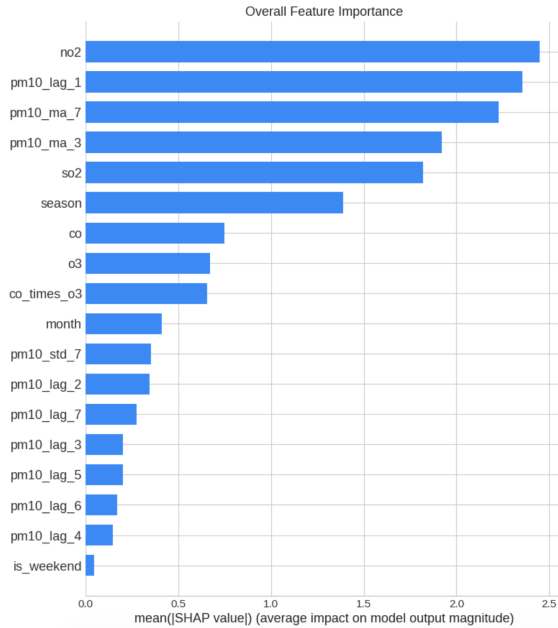


Fig. 7. Global feature importance ranking mapped via average absolute Shapley values (*TreeSHAP*).

Local dependency analysis validates the integration of the composite proxy `co_times_o3`. High coordinate interactions map directly to positive Shapley attributions during periods of low precipitation. This confirms that the Random Forest Regressive model successfully leverages this feature to identify photolytic oxidation states and capture secondary pollution trends during seasonal transitions, achieving algorithmic transparency without sacrificing accuracy.

Furthermore, computational complexity evaluations confirm the practical feasibility of this framework on standard servers. The full network completes its training pass within 16.89 seconds, while requiring an inference latency of only 0.0152 seconds per daily record. This brief processing time supports the feasibility of deploying the system for automated, next-day air quality updates.

IV. CONCLUSION

This study analyzed and evaluated a sequential hybrid RF-ARIMA forecasting configuration based on a 15-feature engineered matrix to project next-day PM_{10} levels in Jakarta. Operating within a DSRM paradigm and justified via pre-development AHP matrices, the architecture successfully separates multi-variate nonlinear interactions from linear temporal dependencies. Based on empirical evaluations on 29,220 daily records across 16 years of data, the proposed model establishes a significant performance improvement, cutting the test RMSE to 7.12 and achieving an R^2 score of 0.89. This represents an explicit error reduction of 22.69% in RMSE and 22.63% in MAE compared to standalone tuned tree models, significantly exceeding the typical 8%–18% improvement range found in global literature. The integration of composite parameters, notably the $CO \times O_3$ photolytic interaction feature, allows the Random Forest Regressor to capture secondary particle formation patterns effectively. Post-hoc TreeSHAP visualizations confirm that the model's internal decision logic aligns with established principles of atmospheric science. Future work may extend this framework by incorporating macro-meteorological attributes, such as wind direction and precipitation volume, and expanding the single-step projection window to support mid-term environmental planning.

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